AIRPORT OPERATIONS PERFORMANCE & FLIGHT DELAY ANALYSIS

A five-year case study of Dubai International Airport (DXB), 2021–2025.

Academic project report

BBA – Business Analytics (IBM)

Year/Semester: Second Year / Third Semester

Tools: Microsoft Excel, SQL and Power BI

Subject: **Business Intelligence (BM 235)** | Teacher: **Mr. Rahul Bisht**

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Study period: Calendar years 2021–2025

Airport: Dubai International Airport (IATA: DXB)

Executive Summary

This project studies the performance of Dubai International Airport (DXB) from 2021 to 2025. It looks at how passenger numbers and aircraft movements changed over five years. It also explains how flight delays can be studied using Excel, SQL and Power BI.

The published figures show that DXB handled 29.1 million passengers in 2021 and 95.2 million in 2025. Aircraft movements increased from 233,375 to 454,800 in the same period. This shows a large increase in airport activity.

Important: Passenger numbers and aircraft movements do not tell us how late flights were. To calculate actual delays, we need each flight's scheduled and actual departure or arrival time. This report does not make up missing delay numbers.

Project Objectives

- Understand the role and growth of Dubai International Airport.
- Compare passenger traffic and aircraft movements from 2021 to 2025.
- Explain what flight delay and on-time performance mean.
- Show simple ways to prepare and analyse flight data using Excel, SQL and Power BI.
- Suggest how airport managers can use the results to improve operations.

What Is Covered in This Report

Section	Main Content
Airport background	Introduction and reason for selecting DXB
Performance analysis	Passenger traffic and aircraft movements, 2021–2025
Flight delay analysis	Simple definitions and calculations
Practical tools	Easy Excel formulas, SQL queries and Power BI measures with sample input and output
Conclusion	Main learning and limits of the available data

1. Introduction to Airport Operations

An airport is a place where many activities happen at the same time. Passengers check in, bags are handled, aircraft are prepared, and flights arrive and depart. Airport operations means managing all these activities safely and smoothly. Good airport operations help flights leave on time, reduce waiting, manage busy periods and give passengers a better experience.

Why airport operations matter

Safety: passengers, staff and aircraft must be protected.

• Time: delays can affect passengers and later flights.

• Service: shorter queues and smooth baggage handling improve the passenger experience.

• Cost: delays can create extra costs for airlines and service providers.

• Planning: airport teams need to manage gates, stands, staff and space.

1.1 About Dubai International Airport Dubai International Airport

(DXB) is the main international airport serving Dubai in the United Arab Emirates. It is operated by Dubai Airports and is one of the world's major international connecting airports. It supports tourism, business and international travel. DXB became the world’s busiest international airport in 2014 according to Dubai Airports. Its passenger traffic later passed the 2019 level in 2023. These facts make DXB a useful airport for studying growth and operations.

1.2 Why DXB was selected

- It handles a very large number of international passengers.
- It has high aircraft movement activity.
- Annual traffic figures are published by Dubai Airports.
- Its growth from 2021 to 2025 gives a useful five-year comparison.
- The topic connects airport management with business analytics.

2. Data and Methodology

This project combines annual airport traffic figures with a separate third-party summary of DXB departure delays. The two datasets cover different periods and measure different things, so they are analyzed separately

2.1 Data used

Data	Source / period	Use in this project
Passengers and aircraft movements	Dubai Airports annual releases, 2021–2025	Measure long-term airport growth
Departure-delay summary	Flight Labs DXB page; sample window 5 Nov 2025–8 Oct 2026; 3,849 departures	Study late-departure share, average lateness and selected airline results
Independent route comparison	Airline Grades: 12,991 departures across 10 routes, Jul 2024–Jul 2026	Context only; not merged with Flight Labs figures

2.2 Simple research steps

- Collect the published annual DXB traffic figures.
- Record the source, date range and sample size for the delay statistics.
- Compare passenger and aircraft movement trends by year.
- Compare delay rates only within the same published sample.

- Check the meaning of each measure before making conclusions.
- Create charts and a dashboard using the published figures.

2.3 What does on time mean?

For this project, a flight is treated as on time when it departs or arrives no more than 15 minutes after its scheduled time. Flight Labs counts a departure as late at 15 minutes past schedule. Different data providers may apply different coverage and definitions.

3. DXB Operational Performance: 2021–2025

The table shows the number of passengers and aircraft movements reported by Dubai Airports. Passenger figures are in millions. The 2024 and 2025 passenger totals are rounded as reported by the airport.

Year	Passengers (million)	Aircraft movements	Passenger growth	Movement growth
2021	29.111	233,375	12.7%	28.1%
2022	66.070	343,339	127.0%	47.0%
2023	86.994	416,405	31.7%	21.3%
2024	92.300	440,300	6.1%	5.7%
2025	95.200	454,800	3.1%	3.3%

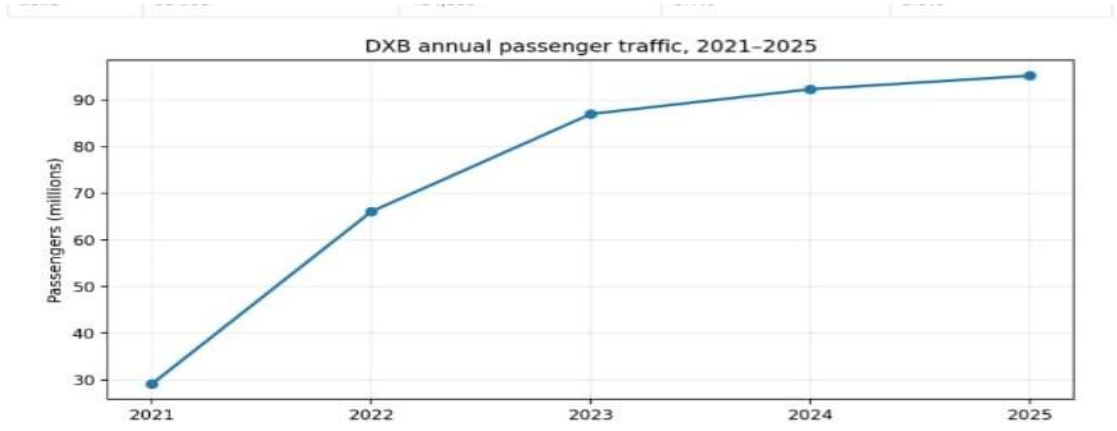


Figure 1. Passenger traffic increased strongly from 2021 to 2023 and continued to grow in 2024 and 2025.

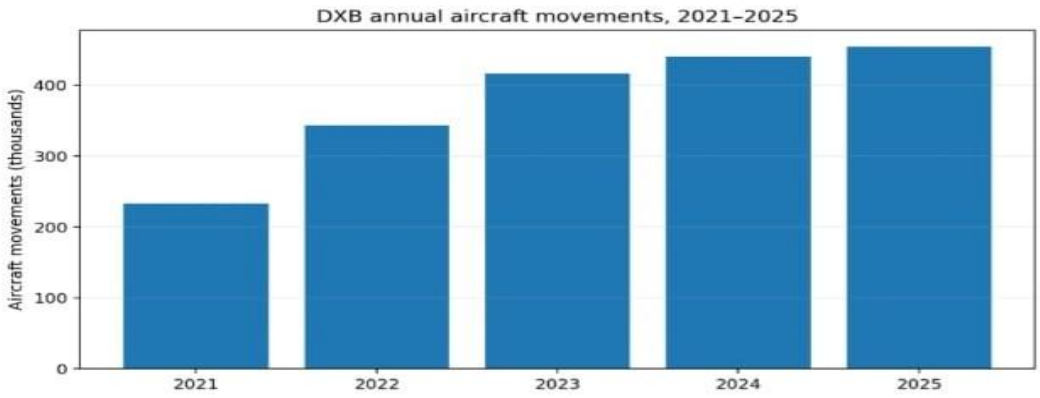


Figure 1. Passenger traffic increased strongly from 2021 to 2023 and continued to grow in 2024 and 2025.

Figure 2. Aircraft movements increased in every year shown.

4. Understanding the Five-Year Trend

2021 **– Recovery begins**

DXB handled 29.1 million passengers and 233,375 aircraft movements. Passenger traffic grew by 12.7% compared with 2020. International travel was beginning to recover.

2022 **– Strong growth**

Passenger traffic increased to 66.1 million and aircraft movements reached 343,339. Passenger traffic grew by 127%. This large increase was partly because travel was recovering from the low levels of the pandemic period.

2023 **– Passenger traffic passes the 2019 level**

DXB handled about 87.0 million passengers and 416,405 aircraft movements. Passenger traffic grew by 31.7%. It was above the 2019 passenger total of 86.4 million.

2024 **– Growth continues**

DXB handled 92.3 million passengers and 440,300 aircraft movements. Passenger traffic increased by 6.1% and aircraft movements by 5.7%.

2025 **– A new record**

DXB handled 95.2 million passengers and 454,800 aircraft movements. Passenger traffic increased by 3.1% compared with 2024. Dubai Airports described 2025 as the busiest year in DXB's history.

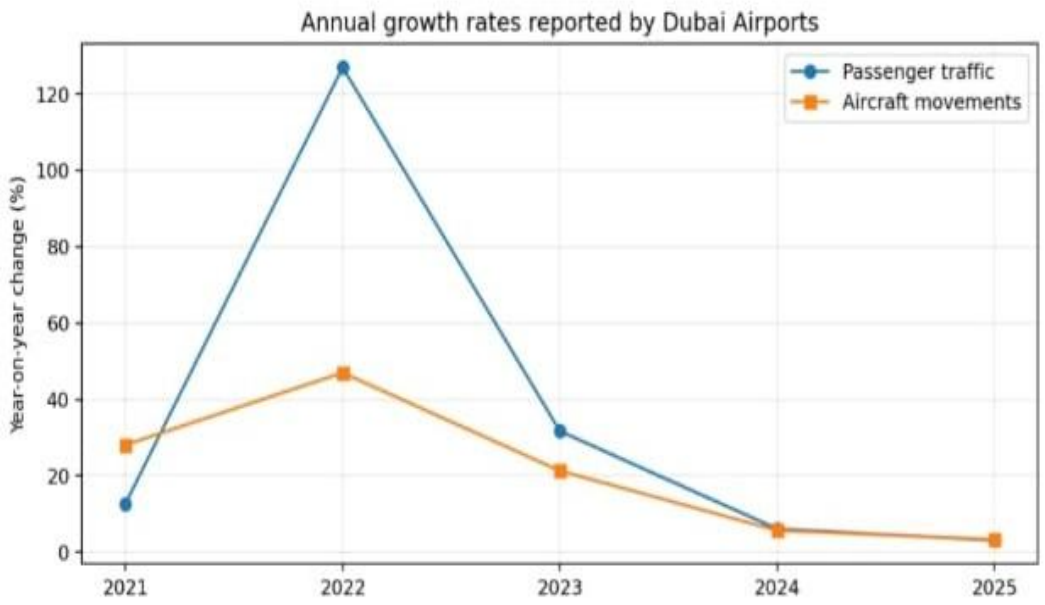


Figure 3. Growth was fastest during the recovery years and became steadier in 2024 and 2025.

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Overall meaning

DXB's passenger numbers and aircraft movements increased greatly between 2021 and 2025. This means the airport had to handle more passengers and aircraft. It is important to plan gates, staff, baggage services and passenger areas carefully. However, higher traffic does not automatically mean more flight delays. Delay data is needed to check that.

5. Flight Delay Analysis

A flight delay is the difference between its scheduled time and actual time. This section uses published DXB departure statistics from FlightLabs rather than invented or demonstration values.

5.1 Published DXB departure sample

Measure	Published result	Meaning	
Departures sampled	3,849	Flights included in the published sample	
Sample period	5 Nov 2025 – 8 Oct 2026	Historical sample window	
Departures 15+ minutes late	85.3%	Share leaving at least 15 minutes after schedule	
Average lateness	42.1 minutes	Average lateness as reported by FlightLabs	

5.2 Airline comparison in the same sample

Airline	Flights sampled	Late departures	Average lateness
Emirates (EK)	1,775	91.2%	43.0 min
flydubai (FZ)	1,107	87.5%	41.3 min
IndiGo (6E)	81	92.6%	43.4 min
Ethiopian (ET)	23	91.3%	38.4 min
Egyptair (MS)	21	95.2%	46.3 min

The airline sample sizes are not equal. Emirates and flydubai have much larger samples than the other airlines shown. A high percentage from a small sample should be interpreted carefully.

5.3 Day and time observations

Flight Labs reports Sunday as the best day in its sample, with 76.5% of departures still recorded as late. The best hour was 23:00–23:59, when 61.6% were late.

"Best" means the lowest late share among the compared day/hour categories; it does not mean most flights were on time.

Source: FlightLabs, "Dubai International Airport (DXB) departure delays". The page states that these are historical aggregates and defines a late departure as 15 minutes or more behind schedule. Data accessed 9 November 2026.

5.4 What the sample suggests

- The published sample shows a high share of departures recorded at least 15 minutes late.
- The reported average lateness is 42.1 minutes.
- Emirates and flydubai account for most of the airline sample shown, so their results have larger sample counts than the other listed carriers.
- The day and hour results suggest that delay monitoring can be broken down by operating period.
- These figures describe the FlightLabs sample only. They should not be treated as a complete measure of every DXB flight.

5.5 Independent comparison and why the figures differ

AirlineGrades separately reports 12,991 DXB departures across 10 routes from July 2024 to July 2026. It reports that 72.78% left within 15 minutes of schedule and 3.03% were cancelled. Its route coverage and date range differ from FlightLabs, so these figures must not be averaged together or combined into one result.

This second source is useful as a cross-check that DXB route-level punctuality statistics are available from another third-party tracker. It is not a full airport-wide census: the page describes 10 routes, and its source notes list public aviation records from several jurisdictions.

5.6 What cannot be concluded

The public summaries do not provide the full flight-by-flight records needed to calculate a complete DXB-wide delay rate for every month, every airline and every year from 2021 to 2025. They also do not prove why an individual flight was delayed. A delay reason should only be stated when a reliable reason code is available.

5.7 How a full dataset should be analyzed

For a complete study, each record should include:

- Flight date
- Airline
- Flight number
- Scheduled departure
- Actual departure
- Scheduled arrival
- Actual arrival
- Flight status
- Where available, a coded delay reason

The analyst can then calculate delay minutes, on-time percentage, average delay, cancellations and monthly or airline comparisons.

Sources:

1. Flight Labs DXB departure-delay summary: https://goflightlabs.com/airport/dxb/delays
2. Airline Grades DXB delays: https://www.airlinegrades.com/airport/dubai-dxb
3. World Bank Development Data Group, Aviation Trends (Aviation Stack data; Dubai validation against OAG aggregates): https://worldbank.github.io/MENA-FCV-economic-monitor/notebooks/aviation/aviation_update.html

6. Excel: Simple Formulas with Input and Output

Excel helps us arrange the flight data and calculate delays. We can use a few simple formulas instead of complicated functions.

Example input table

Flight	Scheduled departure	Actual departure
EK101	**10:00**	**10:20**
FZ202	**11:00**	**11:05**
EK303	**12:00**	**12:30**

Formula 1 – Calculate delay in minutes

=(C2-B2) *1440

Input: Scheduled time = 10:00; Actual time = 10:20

Output: 20 minutes late.

Formula 2 – Mark a flight on time or delayed

=IF (D2<=15,"On Time", “Delayed")

Input: Delay = 20 minutes

Output: Delayed.

Formula 3 – Find average delay

=AVERAGE (D2:D4)

Input: Delays = 20, 5 and 30

Output: 18.33 minutes average delay.

7. SQL: Simple Queries with Input and Output

SQL is used to ask questions from a table of flight records. Below are three basic examples. The sample table is only for explaining how the query works; these are not actual DXB delay results.

Sample input table: flights

flight_id	airline	delay_min	status
1	EK	20	Completed
2	FZ	5	Completed
3	EK	30	Completed
4	FZ	—	Cancelled

Query 1 – Show all completed flights

```
SELECT *
FROM flights
WHERE status = 'Completed';
```

Output: Rows for flights 1, 2 and 3 are shown. The cancelled flight is not included.

Query 2 – Find the average delay

```
SELECT AVG(delay_min) AS average_delay
FROM flights
WHERE status = 'Completed';
```

Output: (20 + 5 + 30) ÷ 3 = 18.33 minutes.

Query 3 – Count flights for each airline

```
SELECT airline, COUNT (*) AS total flights
FROM flights
GROUP BY airline;
```

Output: EK = 2 flights; FZ = 2 flights. This count includes the cancelled sample flight.

8. Power BI: Simple Measures with Input and Output

Power BI turns data into charts and dashboard cards. We can create simple measures and show the result visually. The examples below use the same small sample data as the SQL section.

Measure 1 – Total number of flights

Total Flights = COUNTROWS(flights)

Input: 4 rows in the sample table.

Output: 4 total flights.

Measure 2 – Average delay for completed flights

Average Delay =CALCULATE (

AVERAGE (flights[delay_min]),

flights[status] = "Completed"

Input: Completed-flight delays are 20, 5 and 30 minutes.

Output: 18.33 minutes.

Measure 3 – Count cancelled flights

Cancelled Flights =

CALCULATE(

COUNTROWS(flights),

flights[status] = "Cancelled"

) Input: One row has status Cancelled.

Output: 1 cancelled flight.

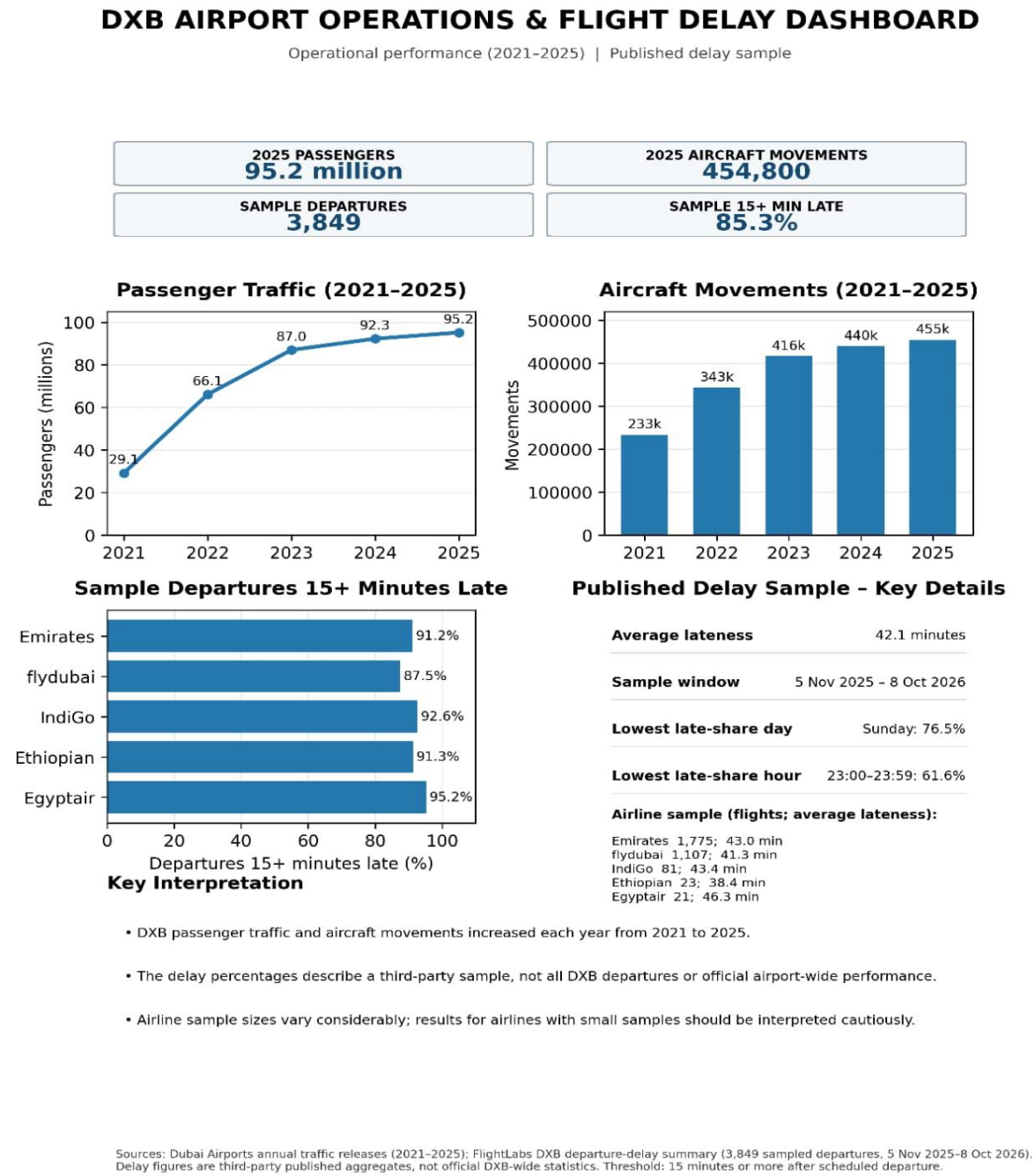
8.1 Suggested Power BI dashboard

Visual	What it shows
Card	Total flights
Card	Average delay in minutes
Card	Number of cancelled flights
Bar chart	Number of flights by airline
Line chart	Average delay by month, when full monthly data is available

The examples above show how a formula produces an output. They use small sample data only to explain the tools. They must not be presented as real DXB performance figures.

8.2 Project Dashboard – DXB Operations at a Glance

This dashboard combines the 2021–2025 annual airport performance figures with the separate FlightLabs departure-delay sample. The delay sample is shown separately so it is not mistaken for a 2021–2025 airport-wide result.



How to read it: The passenger and aircraft charts show the five-year airport growth trend. The delay cards and airline chart show the published third-party sample for 5 November 2025–8 October 2026. The airline counts vary, so small samples should be treated cautiously.

Dashboard source note: Dubai Airports annual traffic figures; FlightLabs DXB departure-delay summary.

9. Findings from the Available Data

Findings from annual DXB airport figures

- Passenger traffic increased from 29.1 million in 2021 to 95.2 million in 2025.

Aircraft movements increased from 233,375 in 2021 to 454,800 in 2025.

- Passenger growth was fastest in 2022 during the recovery in international travel.
- Passenger traffic passed the 2019 level in 2023.
- Growth continued in 2024 and 2025, but at a slower rate than in the recovery years.

Findings from the published delay sample

- FlightLabs reports 3,849 sampled DXB departures from 5 November 2025 to 8 October 2026.
- In this sample, 85.3% departed at least 15 minutes late.
- Average lateness was reported as 42.1 minutes.
- The sample reports different results by airline, day and hour.
- These are third-party sample aggregates, not official DXB-wide annual delay statistics.

What still needs a complete flight-level dataset

- A complete delay trend for each year from 2021 to 2025.
- Monthly delay percentages covering all DXB departures.
- Consistent airline comparisons with adequate sample sizes.

- A complete cancellation and diversion analysis.
- Verified reasons for individual flight delays.

Main interpretation

DXB's annual traffic figures show strong growth. The separate third-party departure sample indicates that delays are an important operational measure to monitor. However, the two datasets cover different periods and have different levels of detail. Good analysis must keep these limits visible and avoid treating a sample as a complete airport record.

10. Recommendations for Airport Operations

These are general recommendations that airport managers can test using reliable operational data. They are not claims that DXB currently has these problems.

Monitor busy hours

Check when the airport has the most flights and passengers, then plan staff and facilities around those times.

Track late incoming aircraft

A late arriving aircraft can affect its next flight. Monitoring this can help teams plan better.

Improve coordination

Airlines, ground staff and airport teams should share updates so that problems can be handled quickly.

Use clear delay reasons

Record delay reasons correctly so managers can understand where problems begin.

Review passenger services

Check baggage handling, security and passenger waiting times along with flight performance.

Check data quality

Make sure flight times are complete and correct before creating reports.

11. Conclusion

This project studied Dubai International Airport's operational growth from 2021 to 2025. The published figures show that passenger traffic and aircraft movements increased greatly during these years. DXB handled 95.2 million passengers and 454,800 aircraft movements in 2025.

The project also explains how flight delays can be measured using scheduled and actual flight times. Simple Excel formulas, SQL queries and Power BI measures show how data can be organized, calculated and presented.

End of the project